

9.4

$$x, y \in C(A^T).$$

$$\text{To show: } AA^T x = AA^T y \Leftrightarrow x = y$$

($\Leftarrow$) trivial

$$(\Rightarrow) \quad AA^T x = AA^T y$$

$$AA^T x - AA^T y = 0$$

$$AA^T \underbrace{(x - y)}_z = 0$$

$$AA^T z = 0 \quad | \cdot z^T$$

$$z^T (AA^T z) = 0$$

$$(z^T A)(A^T z) = 0$$

$$(A^T z)^T (A^T z) = 0$$

$$\|A^T z\|^2 = 0$$

$$A^T z = 0$$

$$z \in C(A^T)$$

$$z \in N(A^T)$$

per
Anfrage

$$C(A^T) \cap N(A^T) = \{0\}$$

$$z = 0$$

$$\Rightarrow x - y = 0$$

$$\Rightarrow x = y$$